

Joel Cruz

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05/08/1995

Experienced Senior Software Engineer with 10+ years in diverse technologies, specializing in Golang and Python. Actively exploring Large Language Models and managing a personal Raspberry Pi cluster. Skilled in writing comprehensive tests, implementing observability, and excelling in teamwork and remote environments. Passionate about leveraging technology to solve complex problems.

PROFESSIONAL EXPERIENCE

LegalZoom,

Senior Full Stack Software Engineer

Glendale, CA

Mar 2023 – Dec 2024

- Utilized Large Language Models (**LLMS**) for accurate detection of filing errors, markedly enhancing the precision of document handling.
- Facilitated the transition of the front end from Server-Side Rendering (**SSR**) with **Django** and React to a Single Page Application (**SPA**), streamlining development by segregating frontend and backend.
- Enhanced legal and business document processing by employing innovative technologies, significantly improving operational efficiency.
- Developed and maintained scalable deployment pipelines using AWS **CDK** for **Beanstalk**, ensuring reliable and efficient application deployment.
- Automated complex workflows and routine tasks with AWS **Lambda**, improving system performance and reducing manual efforts.
- Implemented **GitHub Actions** for continuous integration, focusing on linting checks. Incorporated Flake8 for **Python** and custom linting rules for JavaScript (**ESLint**) to uphold high coding standards.
- Integrated CI/CD pipelines to streamline **code review** processes, maintaining consistency and quality across the development lifecycle.
- Engineered a dynamic workflow system for efficient file and document processing, including creating an on-the-fly PDF scanner to identify and mitigate risks from dangerous files.
- Conducted extensive security research, implemented One-Time Password (**OTP**) features, and designed a comprehensive audit system for secure logging of access to encrypted documents.
- Developed an advanced document indexing and search system utilizing **LLMS**, **RAG** (Retrieval-Augmented Generation), **Elasticsearch**, and **LangChain** to support complex queries and enhance data retrieval.
- Leveraged OpenAI APIs to automate code reviews, generate concise summaries for pull requests, and create test scenarios for AWS Lambda functions, workflows, and new features using **PyTest**.
- Contributed to the development stack with **React** for the frontend and **Django** for the backend, facilitating the creation of responsive and dynamic web applications.

LightStep,

Senior Integration Engineer

San Francisco, CA

Apr 2020 – Feb 2023

- Developed custom data collectors for metrics, logs, and traces using **OpenTelemetry** Collectors, enhancing system **observability** and scalability. Leveraged **Fluentd** and **Jaeger** for comprehensive data management and analysis.

- Orchestrated dynamic Terraform dashboards in **Grafana**, **Datadog**, and Elastic, optimizing infrastructure management and deployment processes. Demonstrated expertise in cloud-based observability and dashboard configuration.
- Innovated automated documentation generators using Golang, Python, and Java. Integrated **ChatGPT**, **OpenAI** APIs, and GitHub **Copilot** to streamline customer integration and service reliability. Pioneered solutions for seamless external service documentation.
- Contributed to OpenTelemetry receivers, demonstrating a commitment to open-source innovation and community collaboration. Enhanced feature sets and functionalities through active code contributions and project involvement.
- Transitioned from a Golang developer to an Integration Engineer, showcasing versatility and a broad skill set. Adapted to new roles and technologies, enhancing team capabilities and project outcomes.
- Utilized APIs (Python, ChatGPT, OpenAI, GitHub Copilot) to convert competitor dashboards (Grafana, Datadog, Elastic) into LightStep's **Terraform** provider files. Improved interoperability and user experience through strategic API use and dashboard conversion.
- Designed and implemented over 100 **metrics** monitoring dashboards from scratch, applying observability principles and software functionalities. Achieved significant improvements in system performance and reliability.
- Automated dashboard creation processes with Python scripts, Terraform, and GitHub Copilot. Enhanced efficiency and streamlined user experiences in dashboard management and deployment.
- Developed tools and examples for load testing data, contributing to improved data handling capabilities and system reliability. Ensured robustness and efficiency in data processing and analysis.
- Managed JIRA tickets using **Agile** methodologies, ensuring organized task management and efficient project execution. Fostered a disciplined and agile project management environment.
- Led weekly **presentations** to showcase dashboard progress and updates, promoting transparency and team collaboration. Encouraged knowledge sharing and project visibility through regular updates.
- Collaborated effectively with team members to align efforts with project goals, driving innovation and project success. Played a key role in team coordination and project alignment.
- Leveraged expertise in GCP, AWS, and Azure observability tools (Google Cloud Monitoring, AWS CloudWatch, Azure Monitor) to enhance **multi-cloud** environment visibility. This resulted in improved system performance and cost optimization.

Megaport,
Senior Software Engineer

San Francisco, CA
Sep 2018 – Feb 2020

- Maintained a critical microservice, ensuring operational efficiency and reliability across the platform.
- Facilitated daily cross-functional meetings with team members in different time zones, especially in Australia, to maintain project alignment and collaborative workflow.
- Utilized the **Prophet** library by Facebook for accurate forecasting of customer usage data, leveraging historical usage patterns to predict future demands.
- Managed and optimized time series data storage, initially using **Cassandra** and later transitioning to **ClickHouse** for improved performance and scalability.
- Designed and executed data pipelines for model training and forecasting, initially employing **Argo CD** for process **orchestration**. To enhance scalability and reliability, migrated to using **NATS** for continuous operation of the Python service, including seamless ClickHouse integration for data management.
- Deployed on AWS **EKS** for production environments, using **Tilt** to streamline project management and development workflows within local Kubernetes setups.
- Authored comprehensive testing frameworks, including unit, integration, and feature tests in Python, ensuring the software's high reliability and flawless functionality.

- Employed a **gRPC** Golang API server to manage tasks, with Python consumers processing these tasks in newer project initiatives.
- Integrated **Prometheus** and **OpenTracing** to monitor and observe application performance, facilitating real-time performance optimization.
- Achieved a significant increase in forecasting capability, improving from 100 predictions per hour to 1000, by shifting from a container-based approach to a more efficient service-based model.
- Explored and researched the integration of GPU technology with **Kubernetes** to further enhance performance and operational efficiency.

WebClues Infotech,
Software Engineer

Jersey City, NJ
Sep 2016 – Aug 2018

- Spearheaded the development and scaling of processes, leveraging **TypeScript**, **Node.js**, and AWS technologies to architect and implement robust front-end applications for direct clients.
- Led a cross-functional team in the creation and delivery of best-in-class solutions, collaborating closely with Product and Engineering teams throughout the development lifecycle.
- Demonstrated proficiency in TypeScript middle-tier development, employing technologies such as Node.js, REST APIs, GraphQL, Axios, PgTyped, Jest, Middy, Swagger, and Zod to build scalable and efficient application components.
- Leveraged TypeScript on the front end to develop **React** functional components, incorporating MUI for enhanced UI/UX and seamlessly integrating **REST** and **GraphQL** APIs for front-end transaction processing.
- Proficiently utilized **Postman** for API testing and development, ensuring API endpoints meet functional requirements and performance benchmarks.
- Integrated multiple applications within AWS, utilizing services such as AWS Cognito, Lambda, API Gateway, CloudWatch, SNS, SQS, S3, and IAM to create robust and scalable architectures.
- Implemented security and data protection measures, ensuring compliance with industry standards and regulations, and collaborated with platform engineering teams to create and manage CI/CD pipelines and infrastructure.
- Interacted with external teams to navigate ambiguous situations and drive optimal solutions based on customer priorities, leveraging strong communication skills and technical expertise to lead cross-functional initiatives.
- Proficiently wrote unit and automation tests, and troubleshooted, debugged, and maintained applications, ensuring high reliability and performance in production environments.

New Jersey City University

Jersey City, NJ
May 2016

Bachelor of Computer Science

- Graduated with a comprehensive understanding of theoretical and practical aspects of computer science.
- Engaged in projects that honed skills in software development, algorithms, and systems design.
- Participated in coding boot camps and hackathons, fostering a collaborative and innovative mindset.
- Completed coursework that included Advanced Programming, Data Structures, Computer Systems, and Software Engineering.
- Contributed to research in distributed computing, resulting in a deeper understanding of networked systems.

SKILLS

Programming Languages

- Python: Advanced proficiency in Python for backend development, AI & ML projects, and data analysis. Skilled in writing clean, efficient, and asynchronous code with asyncio. Proficient in Django, Flask, and FastAPI for web development, and experienced with Pandas and NumPy for data manipulation and numerical computation.
- Go: Advanced knowledge in Go for building high-performance, scalable backend systems.
- JavaScript & TypeScript: Skilled in modern JavaScript (ES6+) and TypeScript for frontend and backend development, ensuring scalable and maintainable code.
- C/C++: Competent in C/C++ for performance-critical systems programming.

Backend Development

- Frameworks & Libraries: Proficient with Echo, Gin (Go), Flask, Django (Python), and FastAPI. Familiar with Django ORM for object-relational mapping and MongoEngine for MongoDB interactions within Django applications. Expertise in real-time communication using Pion WebRTC.
- API Development: Skilled in developing RESTful APIs using Flask and Django, ensuring scalability and performance.

Frontend Development

- JavaScript Frameworks: Skilled in React for building single-page applications, React Native for cross-platform mobile app development, and Next.js for server-side rendering and static site generation.

Databases & Data Management

- Relational Databases: Extensive experience with PostgreSQL, including TimeScaleDB, PostGIS, and MySQL. Advanced knowledge in managing database interactions with Django ORM.
- NoSQL & Search Engines: Proficient with MongoDB (MongoEngine), Elasticsearch, Neo4j, and Redis. Familiar with Clickhouse for analytics and large datasets.

Artificial Intelligence & Machine Learning

- Frameworks & Libraries: Proficient in PyTorch for deep learning applications and familiar with OpenAI's GPT and RAG for generative tasks. Competent with Facebook Prophet for time series forecasting.

Development Operations (DevOps) & CI/CD

- Containerization & Orchestration: Experienced with Docker, Kubernetes, and Docker Swarm. Proficient in managing deployments with Helm charts and using Longhorn for volume storage.
- CI/CD Tools & Version Control: Skilled in GitLab CI/CD, GitHub Actions, and well-versed in Git for efficient version control and automation of development pipelines.

Infrastructure & Configuration Management

- IaC & Automation: Knowledgeable in Ansible and Terraform for infrastructure provisioning and management.

Tools & Integrated Development Environments (IDEs)

- IDEs: Proficient in the JetBrains suite and Visual Studio Code, enhancing development workflows and productivity.

Operating Systems

- Platforms: Comfortable with Linux, Unix, and macOS environments, adept at utilizing their capabilities for development and deployment.